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SARDAR PATEL UNIVERSITY
BCA Semester-6 External Examination [NC] – April-2026
US06CBCA53 || Computer Networks



Date : 04-04-2026

Time : 10.00 a.m. – 01.00 p.m.

Total : 70 Marks

Q-1 Select the appropriate option for the following questions:**10**

- 1) _____ have a single communication channel that is shared by all the machines on the network.
 - a) Point-to-Point network.
 - b) People network
 - c) Broadcast network.
 - d) Specialized network
- 2) _____ consists of two distinct components: transmission lines & switching elements.
 - a) Host
 - b) Subnet
 - c) Intermediate System
 - d) Server
- 3) A _____ cable consists of two conductors each with its own plastic insulation, twisted together.
 - a) Twisted-pair
 - b) Coaxial cables
 - c) Fiber Optical
 - d) None of these
- 4) _____ is the set of techniques that allows the simultaneous transmission of multiple signals across a single data link.
 - a). Multiplexing
 - b). De-multiplexing
 - c). CDMA
 - d).None of these.
- 5) A set of layers and protocols is called _____.
 - a) network architecture
 - b) Protocol stack
 - c) Computer architecture
 - d) Layers stack
- 6) Bit representation is the responsibilities of _____.
 - a) Physical layer
 - b) Session layer
 - c) Transport layer
 - d) Application layer
- 7) Which of the following topology requires maximum cabling?
 - a) Bus
 - b) Star
 - c) Ring
 - d) Mesh
- 8) A Router operates at _____ layer.
 - a) physical
 - b) datalink
 - c) network
 - d) all of these
- 9) The connection of two or more networks is called an _____.
 - a) computer network
 - b) internet
 - c) internetwork
 - d) local Area Network
- 10) _____ is an analog multiplexing technique that combines analog signals.
 - a).FDM
 - b).TDM
 - c).WDM
 - d).None of these.

Q-2 Give short answers of the following questions: (Attempt TEN out of TWELVE)**20**

- 1) Define terms : Computer network, Data Rate
- 2) What is computer network? What are the three categories of computer network?

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(P.T.O.)

- 3) List advantages of computer network.
- 4) Explain Magnetic media in short.
- 5) Explain serial transmission in short.
- 6) What do you mean by Multiplexing? List out various Techniques.
- 7) List design issues for layers.
- 8) Explain responsibilities of session layer.
- 9) Explain orbit.
- 10) Explain types of hub.
- 11) List various topologies.
- 12) Explain Repeater

- Q-3 A Explain classification of computer Network. 5
- B Explain WAN in detail. 5

OR

- Q-3 A Explain LAN in detail. 5
- B Explain MAN in detail. 5

- Q-4 A Explain Co-axial cable in detail. 5
- B Explain Twisted pair cable in detail. 5

OR

- Q-4 A Explain FDM in detail. 5
- B Explain Synchronous TDM in detail. 5

- Q-5 A Draw diagram and explain Physical and Data link layers of OSI model. 5
- B Write a note on satellite Network. 5

OR

- Q-5 A Write a note on protocol hierarchies. 5
- B Explain any two categories of satellite in detail. 5

- Q-6 Explain Bus and Star topology in detail. 10

OR

- Q-6 Explain Hub and Repeater in detail. 10

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